DATA TYPE + SYMBOL KEY

TRIGGER Classify a measurement or choose a test.
STEPS 1. Measured on a numerical scale? Continuous: height/BP; discrete: number of tender joints. 2. Categories? Ordered: cancer stage/Apgar; nominal: country/ZIP/event. Numbered categories need not have equal gaps.
SYMBOLS X: random measurement; x_i: observation; $\bar{x}$: sample mean; μ: population mean; s, σ: sample/population SD; s^2, σ^2: variances; n: sample count; ν: df; α: significance level; f: mass/density; F: CDF.
WRITE Height: interval-continuous; ZIP: categorical-nominal.
TRAP A numerical label is not automatically a measured amount.

RAW LIST → MEAN, VARIANCE, SD, SEM

TRIGGER A list of measurements; estimate center, spread, or error in the estimated mean.
Heights: $65, 69, 59, 64, 62, 67, 68, 62, 64, 65$.

1. $$\sum x_i = 645, \quad n = 10$$
$$\bar{x} = \frac{\sum x_i}{n} = \frac{645}{10} = 64.5$$

2.

x_i	65	69	59	64	62
$x_i - 64.5$	.5	4.5	−5.5	−.5	−2.5
$(x_i - 64.5)^2$	.25	20.25	30.25	.25	6.25

x_i	67	68	62	64	65
$x_i - 64.5$	2.5	3.5	−2.5	−.5	.5
$(x_i - 64.5)^2$	6.25	12.25	6.25	.25	.25

3. $$\sum (x_i - \bar{x})^2 = 82.5$$
$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{82.5}{9} = 9.16667$$
4. $$s = \sqrt{s^2} = \sqrt{9.16667} = 3.02765$$
5. $$\text{SEM} = \frac{s}{\sqrt{n}} = \frac{3.02765}{\sqrt{10}} = .957427$$

Equivalent squared-sum shortcut:
6. $$\sum x_i^2 - \frac{(\sum x_i)^2}{n} = 41685 - 645^2/10 = 82.5$$
Full-population formulas:
7. $$\mu = \frac{\sum x_i}{N}, \quad \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

WRITE Estimated population mean: 64.5 inches; individual SD: 3.02765 inches; mean-estimation error (SEM): .957427 inch.
TRAP SD describes individuals; SEM describes sample means. Entire population measured? Use N and divide squared distances by N, not $n-1$.

SAMPLE PERCENTILE + DATA DISPLAY

TRIGGER A sorted/raw list and a requested percentile, median, IQR or shape.
Sorted $3, 7, 8, 9, 14$; 30th percentile:
1. $$r = 1 + \frac{p(n-1)}{100} = 1 + \frac{30(5-1)}{100} = 2.2 = I + d$$
2. $$x_p = x_{(I)} + d[x_{(I+1)} - x_{(I)}]$$
$$x_{30} = 7 + .2(8 - 7) = 7.2$$

Median: middle value for odd count; average two middle values for even count. IQR: Q3 minus Q1. Range: minimum to maximum; width: maximum minus minimum.
WRITE The 30th sample percentile is 7.2. A roughly straight Normal Q–Q plot supports a Normal shape.
TRAP Histogram: bin counts; box: Q1–Q3 with median; whisker rule: 1.5×IQR; violin width: concentration. Use the lecture rank rule, not an n+1 rule.

QUARTILES + IQR: LOCATE, INTERPOLATE, SUBTRACT

TRIGGER A data list; asks for quartiles, middle spread or a box plot.
Sorted heights: $59, 62, 62, 64, 64, 65, 65, 67, 68, 69$; $n = 10$.
1. $$r = 1 + \frac{p(n-1)}{100}: \quad r_{25} = 1 + .25(9) = 3.25$$
2. $$Q_1 = x_{(3)} + .25[x_{(4)} - x_{(3)}]$$
$$= 62 + .25(64 - 62) = 62.5$$
3. $$r_{50} = 1 + .50(9) = 5.5$$
4. $$\text{median} = 64 + .5(65 - 64) = 64.5$$
5. $$r_{75} = 1 + .75(9) = 7.75$$
6. $$Q_3 = 65 + .75(67 - 65) = 66.5$$
7. $$\text{IQR} = Q_3 - Q_1 = 66.5 - 62.5 = 4$$

WRITE Middle half of the heights: 62.5–66.5 inches; median 64.5 inches.
TRAP Use the fractional rank to move between two sorted observations; do not round it to the nearest rank.

PROBABILITY TABLE → MOMENTS + CDF

TRIGGER Separate possible values with a probability attached to each.
1. $$f(x) \geq 0, \quad \sum_x f(x) = .55 + .15 + .10 + .20 = 1$$

x	−1	2	3	7
$f(x)$	.55	.15	.10	.20
$xf(x)$	−.55	.30	.30	1.40
$x^2 f(x)$	.55	.60	.90	9.80

2. $$\mu = E[X] = \sum xf(x) = -.55 + .30 + .30 + 1.40 = 1.45$$
3. $$E[X^2] = \sum_x x^2 f(x) = .55 + .60 + .90 + 9.80 = 11.85$$
4. $$\text{Var}(X) = E[X^2] - \mu^2 = 11.85 - 1.45^2 = 9.74750$$
5. $$\text{Var}(X) = \sum_x (x - \mu)^2 f(x)$$
$$E[(X-\mu)^2] = .55(-2.45)^2 + .15(.55)^2 + .10(1.55)^2 + .20(5.55)^2 = 9.74750$$
6. $$F(t) = \sum_{x \leq t} f(x)$$
$$F(t) = \begin{cases} 0 & t < -1 \\ .55 & -1 \leq t < 2 \\ .70 & 2 \leq t < 3 \\ .80 & 3 \leq t < 7 \\ 1 & t \geq 7 \end{cases}$$
7. $$P(2 \leq X < 7) = F(7^-) - F(2^-) = .80 - .55 = .25$$

WRITE Mean 1.45; variance 9.74750; probability of 2 up to but excluding 7 is .25.
TRAP $F(a^-)$ excludes mass at a. $P(X > a) = 1 - F(a)$; $P(X \geq a) = 1 - F(a^-)$. A total of .9 is invalid.

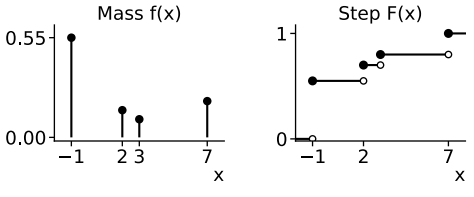
LOADED DIE → WEIGHT BY THE STATED MASSES

TRIGGER A die has unequal probabilities; asks a chance, mean or variance.
Loaded die: $f(1) = .20$, $f(2), \ldots, f(6) = .16$.
1. $$\sum_x f(x) = .20 + 5(.16) = 1$$
$$P(X \leq 3) = F(3) = \sum_{x \leq 3} f(x) = .20 + .16 + .16 = .52$$
2. $$E[X] = \sum_x xf(x) = 1(.20) + (2 + 3 + 4 + 5 + 6)(.16)$$
$$= 3.4$$
3. $$E[X^r] = \sum_x x^r f(x)$$
$$E[X^2] = .20 + (4 + 9 + 16 + 25 + 36)(.16) = 14.6$$
4. $$\text{Var}(X) = 14.6 - 3.4^2 = 3.04$$

WRITE Loaded-die mean 3.4, variance 3.04.
TRAP Counting favorable faces out of six is valid only when all faces are equally likely.

DRAW A DISCRETE MASS FUNCTION + CDF

TRIGGER Sketch a discrete probability/distribution function.
1. Mark each allowed x; draw its mass as a vertical stem.
2. CDF:
$$F(t) = \sum_{x \leq t} f(x)$$
Hold each level flat, then jump by the next mass.
3. Fill the new/right level; leave the old/left level open.



WORKED / WRITE At $x = 2$, the CDF jumps from .55 to .70. Thus $F(2) = .70$, while $P(X < 2) = .55$.
TRAP Only the CDF must never decrease.

DISCRETE CDF: INCLUDE OR EXCLUDE THE ENDPOINT

TRIGGER A discrete probability with a strict or inclusive inequality.
Use the probability table above. $F(a^-)$ is the total just before a.
1. $$F(2^-) = .55, \quad F(2) = .55 + .15 = .70$$
2. $$P(X > a) = 1 - F(a): \quad P(X > 2) = 1 - .70 = .30$$
3. $$P(X \geq a) = 1 - F(a^-): \quad P(X \geq 2) = 1 - .55 = .45$$
4. $$P(a < X \leq b) = F(b) - F(a)$$
$$P(2 < X \leq 7) = F(7) - F(2) = 1 - .70 = .30$$
5. $$P(-1 < X < 3) = F(3^-) - F(-1)$$
$$= .70 - .55 = .15$$

WRITE Including 2 adds its mass .15: the right-tail chance changes from .30 to .45.
TRAP A discrete point can have positive probability. The continuous rule that a single point has zero probability does not apply.

DENSITY → AREA, CDF, MEAN, VARIANCE, MEDIAN

TRIGGER A density formula over a continuous interval; probabilities are areas.
Analysis time: $f(x) = x/2$, $0 \leq x \leq 2$, zero elsewhere.
1. $$\int_a^b f(x)\,dx = \int_0^2 \frac{x}{2}\,dx = \left[\frac{x^2}{4}\right]_0^2 = 1$$
2. $$F(t) = \int_a^t f(u)\,du$$
$$F(t) = \int_0^t u/2\,du = t^2/4 \quad (0 \leq t \leq 2)$$

CDF is zero below 0 and one above 2.
3. $$P(X < 1) = F(1) = \frac{1^2}{4} = .25$$
4. $$P(X > t) = 1 - F(t)$$
$$P(X > 1.5) = 1 - \frac{1.5^2}{4} = .4375$$
5. $$E[X] = \int_a^b xf(x)\,dx$$
$$\mu = \int_0^2 x(x/2)\,dx = [x^3/6]_0^2 = 4/3$$
6. $$E[X^r] = \int_a^b x^r f(x)\,dx \quad (r = 2)$$
$$E[X^2] = \int_0^2 x^2(x/2)\,dx = [x^4/8]_0^2 = 2$$
7. $$\text{Var}(X) = E[X^2] - \mu^2 = 2 - (4/3)^2 = 2/9$$
8. $$F(q) = .5: \quad q^2/4 = .5 \Rightarrow q = \sqrt{2} = 1.41421$$

WRITE Mean time 1.33333 hours; median 1.41421 hours; variance .222222 hours².
TRAP An exact continuous point has probability zero. Density height can exceed one; total area cannot. Median is not necessarily the interval midpoint.

PIECEWISE DENSITY → CARRY EARLIER AREA

TRIGGER The density changes its expression at a boundary.
$f(x) = x$ on 0–1; $f(x) = 2 - x$ on 1–2.
1. $$F(1) = \int_0^1 u\,du = 1/2$$
2. $$F(t) = \int_a^s f_1(u)\,du + \int_s^t f_2(u)\,du \quad (t > s)$$
$$F(t) = \tfrac{1}{2} + \int_1^t (2 - u)\,du$$
$$= \tfrac{1}{2} + [2u - u^2/2]_1^t = 2t - t^2/2 - 1$$
3. $$F(t) = \begin{cases} 0 & t < 0 \\ t^2/2 & 0 \leq t \leq 1 \\ 2t - t^2/2 - 1 & 1 < t \leq 2 \\ 1 & t > 2 \end{cases}$$

WRITE Both interior formulas give .5 at 1; the final value is 1 at 2.
TRAP Do not restart the accumulated probability at zero when the density formula changes.

SEQUENCE OF EVENTS → UPDATE WHAT REMAINS

TRIGGER Coins/cards or sequential trials; asks the chance of a combined outcome.
Two kings without replacement: after a king, 3 kings remain among 51 cards.
1. $$P(A \cap B) = P(A)P(B \mid A)$$
$$P(\text{two kings}) = \frac{4}{52}\frac{3}{51} = .00452489$$

Three fair tosses, exactly two heads: HHT, HTH, THH.
2. $$P = 3/8 = .375$$

WRITE The chance of two kings is .00452489.
TRAP Independent events retain the original second chance. Past heads do not alter the next fair toss.

INTEGRALS: SUBSTITUTE BOTH BOUNDS

Analysis time: $f(x) = x/2$ on 0–2. An antiderivative G is a function whose derivative is f.
AREA Integrate the density; insert upper and lower bounds:
1. $$P(l < X < u) = F(u) - F(l) = \int_l^u f(x)\,dx$$
$$= G(u) - G(l)$$
$$P(1.5 < X \leq 2) = \left[\frac{x^2}{4}\right]_{1.5}^2$$
$$= \frac{2^2}{4} - \frac{1.5^2}{4} = 1 - .5625 = .4375$$

MEAN Multiply the density by x before integrating:
$$E[X] = \int_0^2 x\frac{x}{2}\,dx = \left[\frac{x^3}{6}\right]_0^2$$
$$= \frac{2^3}{6} - \frac{0^3}{6} = \frac{8}{6} = \frac{4}{3}$$

UNKNOWN CONSTANT / ENDPOINT → TOTAL AREA ONE

TRIGGER A density contains an unknown normalizing constant or support endpoint.

$f(x) = .1 + x$ for $0 < x \le c$; zero otherwise.

1. $$\int_{\text{support}} f(x)\,dx = 1$$
$$1 = \int_0^c (.1 + x)\,dx = [.1x + x^2/2]_0^c$$
2. $$.1c + c^2/2 = 1 \Rightarrow c^2 + .2c - 2 = 0$$
3. $$Ac^2 + Bc + D = 0: \quad c = \frac{-B \pm \sqrt{B^2 - 4AD}}{2A}$$
$$c = \frac{-.2 + \sqrt{.2^2 - 4(1)(-2)}}{2} = 1.31774$$
4. $$F(t) = \begin{cases} 0 & t \le 0 \\ .1t + t^2/2 & 0 < t \le c \\ 1 & t > c \end{cases}$$
5. $$P(X > .5) = 1 - [.1(.5) + .5^2/2] = .825$$
6. $$P(0 < X < .6) = .1(.6) + .6^2/2 = .24$$
7. $$P(X = .5) = 0$$

For the mean, multiply by x first:

8. $$\mu = \int_0^c x(.1 + x)\,dx = [.05x^2 + x^3/3]_0^c$$
$$= .05c^2 + c^3/3 = .849556$$
9. $$E[2X + 1] = 2\mu + 1 = 2(.849556) + 1 = 2.69911$$

WRITE The support ends at 1.31774; mean .849556; chance above .5 is .825.

TRAP Keep the unrounded root in subsequent work. A negative root cannot be this upper endpoint.

UNIFORM → WIDTH DIVIDED BY FULL WIDTH

TRIGGER A constant density over a fixed range.

1. $$f(x) = \frac{1}{b-a} = \frac{1}{1-0} = 1 \quad (0 < x < 1)$$
$$F(t) = \frac{t-a}{b-a} = \frac{t-0}{1-0} = t \quad (0 < t < 1)$$
$$P(l < X < u) = \frac{u-l}{b-a} = u - l \quad (0 \le l < u \le 1)$$

Standard Uniform on 0–1:

2. $$E[X] = \int_0^1 x\,dx = [x^2/2]_0^1 = 1/2$$
3. $$E[X^2] = \int_0^1 x^2\,dx = [x^3/3]_0^1 = 1/3$$
4. $$\text{Var}(X) = 1/3 - (1/2)^2 = 1/12$$
5. $$\mu = (a+b)/2, \quad \sigma^2 = (b-a)^2/12$$

WRITE Standard Uniform mean .5 and variance .0833333.

TRAP CDF is zero before a, a rising line inside, one after b.

DENSITY COEFFICIENT

TRIGGER Unknown multiplier; fixed support.

Analysis-time density on 0–2:

1. $$1 = \int_a^b kg(x)\,dx = k\int_a^b g(x)\,dx$$
$$1 = \int_0^2 kx\,dx = k[x^2/2]_0^2 = 2k$$
2. $$k = 1/2 \Rightarrow f(x) = x/2$$

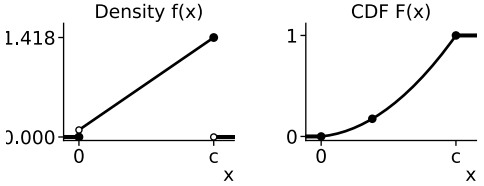
WRITE Density: x/2 on 0–2.

TRAP Normalize over the full support.

DRAW THE CONTINUOUS PDF + CDF

TRIGGER Sketch the density/distribution function after solving for the support.

1. Mark 0 and $c \approx 1.318$ on x; label the vertical axis f or F.
2. $$f(x) = .1 + x$$
PDF: straight line from open $(0, .1)$ to filled $(c, 1.418)$; zero outside. The point $(0, 0)$ is filled.
3. $$F(x) = .1x + x^2/2$$
CDF: through $(0,0), (.5, .175), (c, 1)$; bend upward, then remain at one.
$$F(.5) = .1(.5) + (.5)^2/2 = .175$$



WRITE CDF is continuous from zero to one. **TRAP** A density jump need not cause a CDF jump.

NORMAL → LEFT / RIGHT / BETWEEN AREA

TRIGGER Normal mean and SD plus a measurement cutoff; $X \sim N(\mu, \sigma^2)$.

Glucose: $\mu = 95, \sigma = 8.7$ mg/dL.

1. $$z_{85} = \frac{x-\mu}{\sigma} = \frac{85-95}{8.7} = -1.14943 \approx -1.15$$
2. $$z_{115} = \frac{x-\mu}{\sigma} = \frac{115-95}{8.7} = 2.29885 \approx 2.30$$
3. $$\Phi(z) = P(Z \le z)$$
$$\Phi(1.15) = .8749, \quad \Phi(2.30) = .9893$$
4. $$P(X < 85) = \Phi(-1.15) = 1 - \Phi(1.15)$$
$$= 1 - .8749 = .1251$$
5. $$P(l < X < u) = \Phi(z_u) - \Phi(z_l)$$
$$P(85 \le X < 115) = .9893 - .1251 = .8642$$
6. $$P(X \ge 115) = 1 - .9893 = .0107$$
7. $$.1251 + .8642 + .0107 = 1$$
8. $$z_0 = (0 - 95)/8.7 = -10.9195$$
9. $$P(X < 0) \approx 0, \quad P(X = 95) = 0$$

TABLE z table: find the complete positive z value in a repeated z/area pair; the adjacent area is cumulative LEFT probability.

WRITE About 86.42% fall between 85 and 115 mg/dL using the table.

TRAP The second Normal parameter is variance; take its square root for SD. Individual: use σ; average of n: use SEM at **Sample means and SEM**.

GIVEN AREA → NORMAL CUTOFF / PERCENTILE

TRIGGER A requested percentile, top-tail fraction, or central range.

Glucose 97.5th percentile / top 2.5%:

1. $$p = .975, \quad z_p = \Phi^{-1}(p) = \Phi^{-1}(.975) = 1.96$$
2. $$x_p = \mu + z_p\sigma = 95 + 1.96(8.7) = 112.052$$

Lower/upper quartiles:

3. $$z_{.25} = -.674490, \quad z_{.75} = .674490$$
4. $$Q_1 = 95 - .674490(8.7) = 89.1319$$
5. $$Q_3 = 95 + .674490(8.7) = 100.868$$

Middle 95% leaves .025 in each tail:

6. $$p_L = (1 - .95)/2 = .025, \quad p_U = 1 - .025 = .975$$
7. $$x_L = 95 - 1.96(8.7) = 77.948$$
8. $$x_U = 95 + 1.96(8.7) = 112.052$$

TABLE Search the z table for .9750 to read 1.96. For a right-tail area, first subtract it from one.

WRITE 97.5% lie below about 112.052 mg/dL; central 95% spans 77.948–112.052.

TRAP A data-list percentile uses the rank rule at **Sample percentiles**, not inverse Normal.

NORMAL PDF / CDF → LABEL MEAN + SCALE

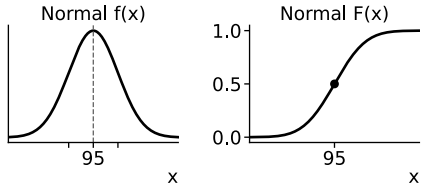
TRIGGER Write or sketch the Normal density/distribution.

1. $$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$
$$f(x) = \frac{1}{8.7\sqrt{2\pi}} e^{-(x-95)^2/(2(8.7)^2)}$$

PDF: symmetric bell at 95; mark 86.3 and 103.7 (one SD each side).

2. $$F(t) = \int_{-\infty}^{t} \frac{1}{\sigma\sqrt{2\pi}} e^{-(u-\mu)^2/(2\sigma^2)}\,du$$
$$F(t) = \int_{-\infty}^{t} \frac{1}{8.7\sqrt{2\pi}} e^{-(u-95)^2/(2(8.7)^2)}\,du$$

CDF: rising S through (95,.5), flattening toward 0 and 1. Insert mean 95 and SD 8.7.



WRITE The integral is sufficient for the CDF.

TRAP CDF height is probability below x; PDF height alone is not probability.

AVERAGE OF n → SAMPLING DISTRIBUTION / SEM

TRIGGER Probability concerns an average. For non-Normal individuals, independent n≥30 is the course guideline for a Normal approximation to the mean.

Impurity: $\mu = 4, \sigma = 1.5, n = 50$. Find $P(3.5 < \bar{X} < 3.8)$.

1. $$E[\bar{X}] = \mu = 4, \quad \text{Var}(\bar{X}) = \sigma^2/n$$
$$\text{SEM} = \frac{\sigma}{\sqrt{n}} = \frac{1.5}{\sqrt{50}} = .212132$$
2. $$z_L = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} = \frac{3.5-4}{.212132} = -2.35702$$
3. $$z_U = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} = \frac{3.8-4}{.212132} = -.942809$$
4. $$P = \Phi(-.942809) - \Phi(-2.35702)$$
$$= .172889 - .00921106 = .163678$$

With the supplied rounded table:

5. $$P = .1736 - .0091 = .1645$$

TABLE Round z to −.94 and −2.36, use symmetry, then subtract. Calculator full-precision result differs slightly from rounded table.

WRITE About 16.4% of such sample averages fall between 3.5 and 3.8 grams.

TRAP Normal individuals give Normal averages for any n. CLT makes the average approximately Normal at large n; it does not make the original observations Normal.

CHANGED VARIABLE / INDEPENDENT SUM → MEAN + VARIANCE

TRIGGER A constant is added, values are rescaled, or independent random variables are added/subtracted.

Subtract 26 from every glucose measurement:

1. $$E[aX + b] = a\mu + b$$
$$E[X - 26] = 1(95) - 26 = 69$$
2. $$\text{Var}(aX + b) = a^2\sigma^2$$
$$\text{Var}(X - 26) = 1^2(8.7)^2 = 75.69, \quad \text{SD} = 8.7$$

Mean uncertainty shrinks because averaging divides a sum by n:

3. $$E[aX + bY + c] = a\mu_X + b\mu_Y + c$$
$$\text{Var}(aX + bY + c) = a^2\sigma_X^2 + b^2\sigma_Y^2$$
$$\text{Var}\left(\frac{\sum X_i}{n}\right) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$
4. $$E[h(X)] = \sum_x h(x)f(x) \quad \text{(discrete)}$$
$$E[h(X)] = \int h(x)f(x)\,dx \quad \text{(continuous)}$$

WRITE A shift changes the mean but not spread; a sum of independent Normal variables is Normal.

TRAP The sum-of-variances rule requires independence. A negative coefficient still contributes positive variance.

ESTIMATES + WHICH SPREAD?

TRIGGER Explain reliability of an estimate, SD versus SEM, or changing sample size.

ESTIMATES Estimate population mean with sample mean and population SD with sample SD. Unbiased means correctly centered over repeated samples. For a Normal population, the sample mean is the minimum-variance mean estimator discussed in class.

Uncertainty in the sample variance:

1. $$\widehat{\text{SE}}(s^2) = s^2\sqrt{\frac{2}{n-1}}$$

Four times the sample size halves SEM:

2. $$\text{SEM}_{4n} = \frac{\sigma}{\sqrt{4n}} = \frac{1}{2}\frac{\sigma}{\sqrt{n}} = \tfrac{1}{2}\text{SEM}_n$$

WRITE A larger sample gives a more precise mean estimate without necessarily reducing variation among individuals. **TRAP** SEM has measurement units; variance has squared units.

READ pnorm / qnorm + lower.tail

TRIGGER A supplied Normal command or its result.

STEPS 1. Identify cutoff versus probability input. 2. Read mean, SD. 3. FALSE means RIGHT tail; default TRUE means LEFT tail.

1. $$\texttt{pnorm}: x \longmapsto F(x)$$

```
pnorm(85,95,8.7)
[1] 0.1251903
pnorm(115,95,8.7,lower.tail=FALSE)
[1] 0.01075671
```

2. $$\texttt{qnorm}: p \longmapsto x_p$$

```
qnorm(.975,95,8.7)
[1] 112.0517
qnorm(.025,95,8.7,lower.tail=FALSE)
[1] 112.0517
```

WRITE Right-tail .025 gives the same cutoff as left-tail .975: about 112.052 mg/dL. **TRAP** The SD argument is not variance; for a sample mean it must be SEM.

ONE SAMPLE vs FIXED VALUE → t TEST

TRIGGER One numerical sample compared with a known/reference population mean.

LDL: reference 115; student $\bar{x} = 103, s = 16, n = 16$.

1. $$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$$ $$H_0 : \mu_{\text{student}} = 115$$
2. $$\text{SEM} = s/\sqrt{n} = 16/\sqrt{16} = 4$$
3. $$\bar{x} - \mu_0 = 103 - 115 = -12$$
4. $$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{-12}{4} = -3$$ $$\nu = n - 1 = 16 - 1 = 15$$
5. $$|t| = 3 > 2.131 \Rightarrow \text{reject } H_0$$

TABLE t table: df row 15; full TWO-TAILED .05 column → 2.131. Use the magnitude for a two-sided alternative.

WRITE At .05, a two-sided one-sample t test supports a different population mean student LDL from 115 mg/dL; the sample mean is lower.

TRAP The reference is a fixed number, not another sample. Do not pool it or give it a sample size.

TWO INDEPENDENT MEANS → POOLED t

TRIGGER Two independent numerical groups; course common-variance assumption; group sizes may differ.

Cognitive scores: $(\bar{x}_1, s_1, n_1) = (49, 8.4, 22)$; $(\bar{x}_2, s_2, n_2) = (57, 9.5, 27)$.

1. $$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 \neq \mu_2$$ $$s_1^2 = 8.4^2 = 70.56, \quad s_2^2 = 9.5^2 = 90.25$$ $$\nu = n_1 + n_2 - 2 = 22 + 27 - 2 = 47$$
2. $$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$ $$s_p^2 = \frac{21(70.56) + 26(90.25)}{47} = 81.4523$$
3. $$\text{SE}_\Delta = \sqrt{s_p^2(1/n_1 + 1/n_2)}$$ $$\text{SE}_\Delta = \sqrt{81.4523(1/22 + 1/27)} = 2.59213$$
4. $$t = \frac{\bar{x}_1 - \bar{x}_2}{\text{SE}_\Delta} = \frac{49 - 57}{2.59213} = -3.08627$$
5. $$|t| = 3.08627 > 2.013 \Rightarrow \text{reject } H_0$$

TABLE t table: df 47 lies between rows 46 and 48; .05 two-tail cutoffs 2.013 and 2.011. The statistic exceeds both.

WRITE At .05, pooled t supports different population mean cognitive scores; the intervention sample mean is higher.

TRAP Do not average variances unless group sizes are equal. A negative t records group order, not a negative p value.

R t OUTPUT → HEADER, df, p

TRIGGER Supplied t output.

STEPS Read header, df and p; compare p with alpha. Height samples:

Default t.test(freshman, senior)

```
  Welch Two Sample t-test
t = -4.5298, df = 15.611, p-value =
0.0003625
```

t.test(..., var.equal = TRUE)

```
  Two Sample t-test
t = -4.5298, df = 18, p-value = 0.0002594
```

READ → t = statistic; df = reference distribution; p = decision. Both p values are below .05.

1. $$p < \alpha \Rightarrow \text{reject } H_0$$ $$.0003625 < .05; \quad .0002594 < .05$$

WRITE Population mean heights differ at .05; the senior sample mean is higher. **TRAP** Use the printed Welch df; do not replace it with pooled df. Hand method: **Pooled t test**.

SEVERAL EQUAL-SIZE MEANS → BALANCED ANOVA

TRIGGER Three or more numerical groups, same number n in each; means and SDs supplied.

Tumors: means $1188, 1412, 1363$; SDs $189, 217, 191$; $m = 3, n = 12$.

1. $$H_0 : \mu_1 = \cdots = \mu_m; \quad H_1 : \text{not all equal}$$ $$s_{\text{wit}}^2 = \frac{\sum_i s_i^2}{m}$$ $$s_{\text{wit}}^2 = (189^2 + 217^2 + 191^2)/3 = 39763.7$$
2. $$\bar{\bar{x}} = \frac{\sum_i \bar{x}_i}{m} = \frac{1188 + 1412 + 1363}{3} = 1321$$
3. $$(\bar{x}_i - 1321) = (-133, 91, 42)$$
4. $$\sum_i (\bar{x}_i - 1321)^2$$ $$= 17689 + 8281 + 1764 = 27734$$
5. $$s_{\bar{x}}^2 = \frac{\sum_i (\bar{x}_i - \bar{\bar{x}})^2}{m - 1} = \frac{27734}{2} = 13867$$
6. $$s_{\text{bet}}^2 = n s_{\bar{x}}^2 = 12(13867) = 166404$$
7. $$F = \frac{s_{\text{bet}}^2}{s_{\text{wit}}^2} = \frac{166404}{39763.7} = 4.18483$$
8. $\nu_n = m - 1 = 2, \quad \nu_d = m(n - 1) = 3(11) = 33$

TABLE F table: numerator df 2 across top; denominator df 33 down side. At upper-tail .05, rows 32 and 34 give 3.29 and 3.28. F exceeds both → reject.

WRITE At .05, ANOVA supports a difference in population mean tumor size among dose groups. It does not identify the differing pair.

TRAP Use variance, not SD, in the ratio. This balanced formula needs equal group sizes; means/SDs are given for the exam.

MEAN-TEST CONDITIONS + DECISION LANGUAGE

TRIGGER State assumptions, explain a result, or choose the correct tail.

STEPS Numerical interval outcome; independent random observations; Normal population(s). Pooled t/ANOVA also assume common population variance. Balanced ANOVA has equal group sizes.

WORKED Student HR by pet preference: cat means 74.2, dog 76.8; SDs 7.3, 9.1; five students per group.

$$s_{\text{wit}}^2 = (7.3^2 + 9.1^2)/2 = 68.05$$ $$\bar{\bar{x}} = (74.2 + 76.8)/2 = 75.5$$ $$s_{\bar{x}}^2 = \frac{(-1.3)^2 + 1.3^2}{2 - 1} = 3.38$$ $$s_{\text{bet}}^2 = 5(3.38) = 16.9, \quad (\nu_n, \nu_d) = (1, 8)$$

1. $F = s_{\text{bet}}^2/s_{\text{wit}}^2 = 16.9/68.05 = .248347 < 5.32$

$$F \text{ or } \chi^2 : \text{ statistic} > \text{upper cutoff} \Rightarrow \text{reject}$$ $$\text{two-sided t: } |t| > t_{\text{crit}} \Rightarrow \text{reject}$$

TABLE F df (1,8), upper .05: 5.32. **WRITE** A population mean HR difference was not established. **TRAP** A p value is an extreme-result probability under the null, not the probability that the null is true; not rejecting does not prove equality.

WHICH PAIRS? → BONFERRONI

TRIGGER Compare a named family of mean pairs at a familywise significance level.

Filgrastim means $3.7, 4.6, 5.2$; $n = 15$ each; given $s_{\text{wit}}^2 = 2.5267, \nu = 42, k = 3$.

1. $$k = \begin{cases} m(m-1)/2 & \text{all pairs} \\ m - 1 & \text{vs control} \end{cases}$$ $$\text{SE}_{ij} = \sqrt{s_{\text{wit}}^2(1/n_i + 1/n_j)}$$ $$\text{SE} = \sqrt{2(2.5267)/15} = .580425$$
2. $$|t_{13}| = \frac{|\bar{x}_1 - \bar{x}_3|}{\text{SE}_{13}} = \frac{1.5}{.580425} = 2.58431$$
3. $$|t_{12}| = (4.6 - 3.7)/.580425 = 1.55059$$
4. $$|t_{23}| = (5.2 - 4.6)/.580425 = 1.03373$$
5. $$\alpha_{\text{pair}} = \alpha_T/k = .05/3 = .0166667$$
6. $$t_{\text{crit}} = 2.49367; \quad 2.58431 > 2.49367$$

TABLE t row 42, TWO-TAIL level .0166667. If absent, TI-84 invT uses LEFT area 1−.05/(2×3) and df 42. The table uses .0166667; only invT uses the left-area conversion.

WRITE With familywise .05 Bonferroni control, low and high dose population means differ; other pair differences were not established.

TRAP Use ANOVA residual df and within variance, not two-group pooled df.

MULTIPLE PAIRS → HOLM: STOP AT FIRST FAILURE

TRIGGER Holm adjustment; supplied t/chi-square statistics or unadjusted p values.

Five t comparisons; $\nu = 46, \alpha_T = .05$.

1. $$\alpha_j = \frac{\alpha_T}{k - j + 1}, \quad j = 1, \ldots, k$$

pair	$\lvert t \rvert$	α_j	t_{crit}
1 : 6	3.3	.0100	2.687
1 : 3	2.9	.0125	2.59973
1 : 5	2.3	.0166667	2.48469
1 : 4	2.2	.0250	
1 : 2	2.1	.0500	

2. $$3.3 > 2.687: \text{ pass}; \quad 2.9 > 2.59973: \text{ pass}$$
3. $$2.3 < 2.48469: \text{ STOP}$$

TABLE t df row 46; each column is the current full TWO-TAILED alpha. Missing .0125/.0166667 columns: invT left area 1−alpha_j/2.

WRITE At familywise .05, G1 differs from G6 and G3; differences from G5, G4 and G2 were not established.

TRAP Order p smallest first, or absolute t largest first when df match. Stop at the first failure; remaining comparisons are not significant.

READ THE SUPPLIED TABLES WITHOUT DOUBLING TAILS

TRIGGER A critical value, p-value range, or missing table entry.

1. **z table:** find z and its neighboring cumulative LEFT area. Negative z: use symmetry.
2. **t table:** row = df; column = total TWO-TAILED alpha. A .05 test uses .05, not .025.
3. **F table:** numerator df along top, denominator df down side; use the requested UPPER tail.
4. **Chi-square:** df down side; upper-tail probability across top. Ordinary chi-square tables are interchangeable for the same df/tail.

WORKED t=2.58630, df 4:

$$t_{.10,4} = 2.132 < 2.58630 < 2.776 = t_{.05,4}$$ $$.05 < p_{\text{two-sided}} < .10$$

Missing df? Read both neighboring rows. Above both cutoffs → reject; below both → do not reject; between them → an exact cutoff or supplied p is needed.

TI-84 Positive t cutoff: invT(1−alpha/2,df). Two-sided p: 2×tcdf(|t|,1E99,df). Normal area: normalcdf(lower,upper,mean,SD); cutoff: invNorm(left area,mean,SD).

TRAP Halve alpha for the invT left-area conversion, not a second time for the supplied two-tailed table.

REAL summary(aov) → WHAT EACH COLUMN ANSWERS

TRIGGER Supplied ANOVA output.

```
            Df Sum Sq Mean Sq F value Pr(>F)
Group        2  779.4   389.7   5.832 0.0106 *
Residuals   19 1269.5    66.8
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

READ → Group Df 2 = numerator df; Residuals Df 19 = denominator df and pairwise t df. **Mean Sq 66.8** = within variance; **F value 5.832** = statistic; **Pr(>F) .0106** = p value.

1. $$F = \frac{\text{MS}_{\text{Group}}}{\text{MS}_{\text{Residual}}} = 5.832 \quad (\text{unrounded inputs})$$

STEPS Compare p with alpha. At .05 reject; at .01 do not reject. For follow-ups, use residual variance and residual df with the actual pair's sample sizes.

If only residual standard error 8.174069 is printed, square it:

$$s_{\text{wit}}^2 = (8.174069)^2 = 66.8154 \approx 66.8$$

WRITE At .05, at least one population mean exhaled nitric-oxide level differs. **TRAP** Residual standard error is an SD, not a variance; a star is not the numerical p value.

ONE EVENT PROPORTION → ESTIMATE + SE

TRIGGER One group with event/no-event outcomes; estimate the event chance and uncertainty.

Death in 8 of 61 patients:

1. $$\hat{p} = m/n = 8/61 = .131148$$
2. $$\widehat{\mathrm{Var}}(X) = \hat{p}(1-\hat{p}) = .113948$$
3. $$\widehat{\mathrm{SD}}(X) = \sqrt{\hat{p}(1-\hat{p})} = \sqrt{.113948} = .337562$$
4. $$\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{.113948}{61}} = .0432203$$

WRITE Estimated death risk .131148; standard error of the estimated proportion .0432203.

TRAP Individual binary SD and SE of the estimated proportion answer different questions. Name the same event in every group.

TWO PROPORTIONS → YATES z

TRIGGER Two groups; each person has event/no event. Same event definition in both groups.

Placebo 18/25 events; aspirin 6/19:

1. $$H_0 : p_1 = p_2, \quad H_1 : p_1 \neq p_2$$
$$\hat{p}_i = m_i/n_i : \quad \hat{p}_1 = 18/25 = .72$$
$$\hat{p}_2 = 6/19 = .315789, \quad \hat{p} = \frac{m_1+m_2}{n_1+n_2} = \frac{24}{44} = .545455$$
2. $$\mathrm{SE}_0 = \sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}$$
$$\mathrm{SE}_0 = \sqrt{.545455(.454545)(1/25+1/19)} = .151547$$
3. $$c_Y = \tfrac{1}{2}(1/n_1 + 1/n_2)$$
$$c_Y = .5(1/25+1/19) = .0463158$$
4. $$D_Y = \max(0, |\hat{p}_1 - \hat{p}_2| - c_Y)$$
$$D_Y = |.72 - .315789| - .0463158 = .357895$$
5. $$|z_Y| = \frac{D_Y}{\mathrm{SE}_0} = \frac{.357895}{.151547} = 2.36161 > 1.960$$

Rounded-proportion format (dioxin): .565, .455; groups 186, 220; exactly 205 female births total.

6. $$\hat{p} = 205/406 = .504926$$
7. $$\mathrm{SE}_0 = \sqrt{.504926(.495074)(1/186+1/220)} = .0498017$$
8. $$c_Y = .5(1/186+1/220) = .00496090$$
9. $$|z_Y| = \frac{.565 - .455 - .00496090}{.0498017} = 2.10915 > 1.960$$

Female-birth proportions differ at .05.

TABLE t table infinity-df row, full .05 two-tail column → 1.960. Reject.

WRITE At .05, corrected two-proportion z supports different population thrombus rates; aspirin's observed rate is lower.

TRAP Always retain Yates for the course's 2×2 comparisons. If proportions are rounded but an exact total event count is supplied, pool that count.

2×2 COUNTS → YATES CHI-SQUARE

TRIGGER Two rows × two outcome columns; asks whether population proportions differ.

1. $$O = \begin{pmatrix} 18 & 7 \\ 6 & 13 \end{pmatrix}, \quad N = 44$$
2. $$E_{ij} = R_i C_j / N$$
$$E = \begin{pmatrix} 25(24)/44 & 25(20)/44 \\ 19(24)/44 & 19(20)/44 \end{pmatrix}$$
3. $$= \begin{pmatrix} 13.6364 & 11.3636 \\ 10.3636 & 8.63636 \end{pmatrix}$$
4. $$\chi^2_Y = \sum_{ij} \frac{\max(0, |O_{ij} - E_{ij}| - .5)^2}{E_{ij}}$$
$$c_{11} = \frac{(|18 - 13.6364| - .5)^2}{13.6364} = 1.09470$$
5. $$\chi^2_Y = 1.09470 + 1.31364 + 1.44039 + 1.72847$$
$$= 5.57719, \quad \nu = (r-1)(c-1) = 1$$

```
  Pearson's Chi-squared test with Yates' continuity correction
X-squared = 5.5772, df = 1, p-value = 0.0182
```

READ → The R header explicitly includes Yates. X-squared is the statistic; p-value is the upper-tail probability.

TABLE Chi-square row 1, upper-tail .05 → 3.841. For a p range: 5.024 (.025) < 5.57719 < 6.635 (.01), so .01

WRITE At .05, corrected chi-square supports different thrombus proportions for placebo and aspirin.

TRAP Use counts, not percentages. Corrected z squared equals this corrected chi-square.

LARGER TABLE → ORDINARY CHI-SQUARE

TRIGGER More than two rows or columns; compare population outcome distributions across groups.

1. $$H_0 : \text{equal outcome proportions across groups}$$
$$O = \begin{pmatrix} 36 & 131 \\ 41 & 132 \\ 75 & 104 \end{pmatrix}, \quad R = (167, 173, 179)$$
2. $$C = (152, 367), \quad N = 519$$
3. $$E_{11} = \frac{R_1 C_1}{N} = \frac{167(152)}{519} = 48.9094$$
4. $$E = \begin{pmatrix} 48.9094 & 118.091 \\ 50.6667 & 122.333 \\ 52.4239 & 126.576 \end{pmatrix}$$
5. $$c_{11} = (36 - 48.9094)^2/48.9094 = 3.40739$$
6. $$\chi^2 = \sum_{ij} (O_{ij} - E_{ij})^2 / E_{ij}$$
$$\chi^2 = 3.40739 + 1.41124 + 1.84430 + .763851$$
$$+9.72230 + 4.02667 = 21.1757, \quad \nu = 2$$
$$\nu = (r-1)(c-1) = (3-1)(2-1) = 2$$

```
  Pearson's Chi-squared test
X-squared = 21.176, df = 2, p-value = 2.522e-05
```

7. $$\texttt{2.522e-05} = .00002522$$

TABLE Chi-square row 2, upper .05 → 5.991. Reject.

WRITE At .05, population outcome distributions differ among the three treatments.

TRAP No Yates for this 3×2 table. Expected counts may be fractional. Follow-up pairs each need a new 2×2 table and new margins.

ONE COUNT LIST vs TARGET PROPORTIONS → GOF

TRIGGER One observed category distribution compared with specified expected percentages/probabilities.

Observed $(255, 125, 90)$, $N = 470$; targets $(.60, .25, .15)$.

1. $$H_0 : p_i = p_{i0} \text{ for all } i$$
$$E_i = N p_{i0} : \quad E = (470(.60), 470(.25), 470(.15))$$
$$= (282, 117.5, 70.5)$$
2. $$\chi^2 = \sum_i (O_i - E_i)^2 / E_i$$
$$\chi^2 = \frac{(255-282)^2}{282} + \frac{(125-117.5)^2}{117.5} + \frac{(90-70.5)^2}{70.5}$$
$$= 2.58511 + .478723 + 5.39362 = 8.45745, \quad \nu = 2$$
$$\nu = k - 1 = 3 - 1 = 2$$

```
  Chi-squared test for given probabilities
X-squared = 8.4574, df = 2, p-value = 0.01457
```

TABLE Chi-square row 2, upper .05 → 5.991. Reject.

WRITE At .05, the population exercise-category distribution differs from the stated earlier pattern.

TRAP Use the specified target probabilities, not the current observed proportions. No 2×2 Yates correction for a one-list goodness-of-fit test.

SUPPLIED PAIRWISE CHI-SQUARE → CONTROL FAMILY

TRIGGER Printed pairwise test statistics; asks which treatments differ from control.

Control comparisons: G1 9.4944; G2 4.6895; G3 18.234, each df 1. Six pairs were printed, but only three belong to this requested family.

1. $$\alpha_j = \alpha_T/(k-j+1)$$
$$k = 3, \quad \alpha_j = .05/3,\ .05/2,\ .05$$
2. $$18.234 > 6.635 > \chi^2_{.0166667,1} : \text{ pass}$$
3. $$9.4944 > \chi^2_{.025,1} = 5.024 : \text{ pass}$$
4. $$4.6895 > \chi^2_{.05,1} = 3.841 : \text{ pass}$$

TABLE Chi-square row 1. The .0166667 column is absent, but exceeding the stricter .01 cutoff 6.635 already proves the first pass.

WRITE At familywise .05, G1, G2 and G3 each differ from control in the population outcome proportion.

TRAP Do not count unrequested treatment-versus-treatment pairs in k. Keep the named outcome and treatment labels in the answer.

PROSPECTIVE RISKS → RR, ARR, NNT / NNH

TRIGGER Treatment/exposure groups followed for an outcome; asks the size of a risk difference or number treated.

Aspirin 6/19 thrombi; placebo 18/25:

1. $$r_T = 6/19 = .315789, \quad r_C = 18/25 = .72$$
$$\mathrm{RR} = r_T/r_C = .315789/.72 = .438596$$
2. $$\mathrm{ARR} = r_C - r_T = .72 - .315789 = .404211$$
3. $$\mathrm{NNT} = 1/\mathrm{ARR} = 1/.404211 = 2.47396 \to 3 \text{ people}$$

Low-base-rate benefit: deaths 10 to 8 per thousand.

4. $$r_C = .010, \quad r_T = .008, \quad \mathrm{RR} = .8$$
5. $$\mathrm{RRR} = 1 - \mathrm{RR} = 1 - .8 = .2$$
$$\mathrm{ARR} = .010 - .008 = .002, \quad \mathrm{NNT} = 1/.002 = 500$$

If NNH=20, harms per one additional avoided death:

6. $$\mathrm{NNH} = 1/(r_T - r_C) \quad \text{when treatment raises harm}$$
$$\mathrm{NNT}/\mathrm{NNH} = 500/20 = 25$$

WRITE Aspirin thrombus risk is .438596 times placebo's; absolute reduction is 40.4211 percentage points; about three treated per additional avoided event.

TRAP Risk uses events/total; odds use events/non-events. A 20% relative reduction is not a twenty-percentage-point absolute reduction. Use decimal risks in reciprocals.

CASES CHOSEN BY DISEASE → ODDS RATIO

TRIGGER People selected as cases/controls by existing outcome, then earlier exposure assessed.

Cases: 50 exposed, 14 unexposed; controls: 43 exposed, 35 unexposed.

1.

	exposed	unexposed
cases	a	b
controls	c	d

$$\mathrm{OR} = \frac{a/b}{c/d} = \frac{ad}{bc} = \frac{50/14}{43/35}$$
$$= \frac{50(35)}{14(43)} = 2.90698$$

WRITE Odds of secondhand-smoke exposure are about 2.90698 times as high among cases as controls.

TRAP Case-control sampling does not supply population disease-risk denominators for RR. Reversing one comparison takes the reciprocal. RR/OR=1 means no association.

STUDY DESIGN → WHAT WAS ASSIGNED / FOLLOWED?

TRIGGER Identify study design, masking, or choose risk versus odds.

1. **Randomized trial:** assign treatment, then follow outcomes. **Cohort:** follow exposure groups without random assignment; both support event risks/RR.
2. **Case-control:** select by outcome, assess earlier exposure; use OR.
3. **Parallel:** different people in treatments.
   Crossover: same people receive both in different periods; stable disease and no carryover required.

Masking: subject; treatment/evaluation team; statistical team. Each hides assignment from a possible source of expectation bias. A sham controls procedure experience/expectation.

WRITE State what was assigned, how people were selected, and which outcome was followed. **TRAP** Association alone does not establish causation; volunteers need not represent every patient.

FOLATE: RISK RATIOS FROM COUNTS

GIVEN NTD is the event. Folate: 4 with NTD, 471 without; no folate: 27 with NTD, 498 without.

1. TOTALS Add both outcomes within each group:

$$n_F = 471 + 4 = 475, \quad n_{NF} = 498 + 27 = 525$$

2. RISKS Divide events by everyone in that group:

$$r_F = \frac{4}{475} = .00842105, \quad r_{NF} = \frac{27}{525} = .0514286$$

3. RATIO For risk without folate relative to risk with folate:

$$\mathrm{RR} = \frac{r_{NF}}{r_F} = \frac{27/525}{4/475} = 6.10714$$

RESULT The no-folate group's NTD risk was about 6.11 times the folate group's risk.

Swapping the groups takes the reciprocal:

$$\mathrm{RR}_{F:NF} = 1/\mathrm{RR}_{NF:F} = .163743$$